

Adélie Lorge

College Station, TX

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Education

Texas A&M University

Expected May 2027

B.S. Chemical Engineering

College Station, TX

Relevant Coursework: Chemical Engineering Thermodynamics (Study abroad), Experimental Research (CHEN 491)

Experience

Undergraduate Student Researcher with Dr. Pushkar Lele — Materials & Fluid Dynamics

Texas A&M University

Sep 2025 – Present

- Designed and executed experimental workflows to evaluate low-cost alternatives to cleanroom-fabricated silicon masters for PDMS microfluidic device production.
- Fabricated reusable microfluidic molds using photocurable optical adhesives and UV-based patterning techniques.
- Characterized replication fidelity, surface behavior, and degradation across repeated PDMS casting cycles using optical microscopy.
- Analyzed fabrication tradeoffs including cost, accessibility, and performance to assess scalability in non-cleanroom environments.
- Collaborated with graduate and PhD researchers on experimental planning, data analysis, and documentation.
- **First-author manuscript in preparation.**

Rocket Modeling Project — CHEN 201 Lab

Texas A&M University

Aug 2024 – Dec 2024

- Built and calibrated Excel/VBA-based rocket flight performance models using experimental data.
- Automated parameter tuning and calculations to improve prediction accuracy and modeling efficiency.
- Integrated experimental engineering measurements with computational analysis to assess model sensitivity and robustness.

European Scouts — Leader

Houston, TX

Aug 2020 – May 2023

- Led and coordinated activities for 40+ children, managing logistics, safety planning, budgeting, and parent communication.
- Operated in a high-responsibility environment requiring leadership, organization, and risk management.

Technical Skills

Programming & Analysis: MATLAB (modeling, data analysis), Excel (advanced formulas, data analysis), VBA (workflow automation)

Experimental & Engineering: Microfluidic fabrication, PDMS casting, optical microscopy, experimental design, data analysis

Safety & Compliance: Laboratory safety certifications

Certifications

- Biosafety Level 2 (BSL-2) Certification — Texas A&M University
- Introduction to Programming with MATLAB — Coursera